1 Oct 20 HW

Oct 20 homework: Proov 3 equation

1.$[p_x, \psi(x)] = -i\bar{h}\frac{\partial\psi}{\partial x}$

2.$[p_x^2, \psi(x)] = -2i\bar{h}\frac{\partial\psi}{\partial x}p_x - \bar{h}^2\frac{\partial^2\psi}{\partial x^2}$

3.$[x, y(p(x))] = i\bar{h}\frac{\partial y(p(x))}{\partial p(x)}$

proov now:

1:$[p_x, \psi]\psi = p_x[\psi^2] - \psi p_x[\psi] = -i\bar{h}2\psi\partial_x\psi + i\bar{h}\psi\partial_x\psi = -i\bar{h}\partial_x\psi\psi$

2:$[p_x^2, \psi] = -\bar{h}^2\partial_x\partial_x\psi^2 + \bar{h}^2\psi\partial_x\partial_x\psi = -2\bar{h}i\partial_x\psi \times p_x\psi - \bar{h}^2\psi\partial_x\partial_x\psi = (-2\bar{h}i\partial_x\psi p_x - \bar{h}^2\partial_x\partial_x\psi)\psi$

3:$[x, y(p(x))]$ and$y(p(x)) = \Sigma_n C_n p(x)^n$ so $\Sigma_n C_n[x, p_x^n] = [x, y(p(x))]$ and $[x, p_x^n]\psi = x \times p_x^n\psi - p_x^n(x\psi) = in\bar{h}p_x^{n-1}\psi$=$\Sigma_n C_n[x, p_x^n] = [x, \Sigma_n c_n p_x^n] = i\bar{h}\Sigma_n C_n n p_x^{n-1}\psi = i\bar{h}\partial_{p_x}y(p_x)\psi$

$p_x^n(x\psi) = p_x^{n-1}(c\psi + x\partial_x\psi) = cp_x^{n-1}\psi + p_x^{n-2}(c\partial\psi + xp_x^2\psi)$ so $= ncp_x^{n-1}\psi + p_x^n\psi$and$c = -i\bar{h}$

$p_x[x\psi] = -i\bar{h}\psi + xp_x\psi$

D O N E .

2 Oct 23HW part2

PROOF:$e^{a\frac{d}{dt}}\psi = \psi(x+a)$

$e^{a\frac{d}{dt}}\psi = \Sigma_n \frac{a^n}{n!}\frac{d^n}{dt^n}\psi; \psi(x+y) = \Sigma_n \frac{\frac{d^n\psi}{dt^n}|_{y=0}}{n!}y^n$consider y=a:$\Sigma_n \frac{\frac{d^n\psi(x)}{dt^n}}{n!}a^n = \Sigma_n \frac{a^n}{n!}\frac{d^n}{dt^n} = e^{a\frac{d}{dt}}\psi$

DONE.

3 Oct 28HW

A:

let $l_\pm = l_x \pm il_y$

PROOF:

(a).$[l_z, l_\pm] = \pm\bar{h}l_\pm$

(b).$l_\pm l_\mp = l^2 - l_z^2 \pm \bar{h}l_z$

(c).$[l_+, l_-] = 2\bar{h}l_z$

proof a:

$[l_z, l_\pm] = [l_z, l_x] \pm i[l_z, l_y] = i\bar{h}l_y \pm i(-i\bar{h}l_x) = i\bar{h}l_y \pm \bar{h}l_x = \pm\bar{h}l_\pm$

proof b:

$l_\pm l_\mp = (l_x \pm il_y)(l_x \mp il_y) = l_x^2 \mp il_xl_y \pm il_yl_x + l_y^2 = l_x^2 + l_y^2 \pm i(l_yl_x - l_xl_y) = l^2 - l_z^2 \pm i[l_y, l_x] = l^2 - l_z^2 \pm \bar{h}l_z$

proof c:

$l_+l_- - l_-l_+ = (l_x + il_y)(l_x - il_y) - (l_x - il_y)(l_x + il_y) = -il_xl_y + il_yl_x - il_xl_y + il_yl_x = 2i[l_y, l_x] = 2\bar{h}l_z$

B:

PROOF:

(a) $p \times l + l \times p = 2i\bar{h}p$

(b)$i\bar{h}(p \times l - l \times p) = [l^2, p]$

proof a:

$(p_1l_2 - p_2l_1)k + (p_2l_3 - p_3l_2)i + (p_3l_1 - p_1l_3)j + (l_1p_2 - l_2p_1)k + (l_2p_3 - l_3p_2)i + (l_3p_1 - l_1p_3)j =$

$([p_1, l_2] + [l_1, p_2])k + ([p_2, l_3] + [l_2, p_3])i + ([p_3, l_1] + l_3, p_1)j$

$(i2\bar{h}p_3)k + (i2\bar{h}p_1)i + (i2\bar{h}p_2)j = i2\bar{h}\boldsymbol{p}$

proof b:

$i\bar{h}(p \times l - l \times p) = (p_1l_2 - p_2l_1 - l_1p_2 + l_2p_1)k + (p_2l_3 - p_3l_2 - l_2p_3 + l_3p_2)i + (p_3l_1 - p_1l_3 - l_3p_1 + l_1p_3)j$

$[l^2, p] = [l^2, p] = [l_x^2 + l_y^2 + l_z^2, p]$

$[l_x^2, p] = l_x[l_x, p] + [l_x, p]l_x = l_x(i\bar{h}(p_zj - p_yk)) + (i\bar{h}(p_zj - p_yk))l_x = i\bar{h}(l_xp_z + p_zl_x)j - i\bar{h}(l_xp_y + p_yl_x)k$

$[l_y^2, p] = l_y[l_y, p] + [l_y, p]l_y = l_y(i\bar{h}(p_xk - p_zi)) + (i\bar{h}(p_xk - p_zi))l_y = i\bar{h}(l_yp_x + p_xl_y)k - i\bar{h}(l_yp_z + p_zl_y)i$

$[l_z^2, p] = l_z[l_z, p] + [l_z, p]l_z = l_zi\bar{h}(p_yi - p_xj) + i\bar{h}(p_yi - p_xj)l_z = i\bar{h}(l_zp_y + p_yl_z)i - i\bar{h}(l_zp_x + p_xl_z)j$

$[l^2, p] = i\bar{h}(l_yp_x + p_xl_y - l_xp_y - p_yl_x)k + i\bar{h}(l_zp_y + p_yl_z - l_yp_z - p_zl_y)i + i\bar{h}(l_xp_z + p_zl_x - l_zp_x - p_xl_z)j$

so we have $i\bar{h}(p \times l - l \times p) = [l^2, p]$

4 Oct 30 HW

$\psi_n = A_ne^{-\alpha^2x^2/2}H_n(\alpha x)$

2.7

(1)

$H_n(\alpha x) = \frac{1}{2\alpha x}(H_{n+1}(\alpha x) + 2nH_{n-1}(\alpha x))$

$x\psi_n = \frac{1}{2\alpha}(H_{n+1}(\alpha x) + 2nH_{n-1}(\alpha x))A_ne^{-\alpha^2x^2/2}$

$A_n = \frac{1}{\sqrt{2n}}A_{n-1}; A_n = \sqrt{2n+2}A_{n+1}$

so we get:$x\psi_n = \frac{1}{2\alpha}(A_nH_{n+1}(\alpha x) + 2nA_nH_{n-1}\alpha x)e^{-\alpha^2x^2/2}$

$\frac{1}{2\alpha}(\sqrt{2(n+1)}A_{n+1}(\alpha x) + 2n\frac{1}{\sqrt{2n}}A_{n-1}H_{n-1}\alpha x)e^{-\alpha^2x^2/2}$

$= \frac{1}{2\alpha}(\sqrt{2(n+1)}\psi_{n+1} + \sqrt{2n}\psi_{n-1})$

(2)

$x^2\psi_n = \frac{1}{2\alpha}(\sqrt{2(n+1)}x\psi_{n+1} + \sqrt{2n}x\psi_{n-1})$

$x^2\psi_n = \frac{1}{2\alpha}(\sqrt{2(n+1)}[\frac{1}{2\alpha}(\sqrt{2(n+2)}\psi_{n+2} + \sqrt{2(n+1)}\psi_n)] + \sqrt{2n}[\frac{1}{2\alpha}(\sqrt{2n}\psi_n + \sqrt{2(n-1)}\psi_{n-2})]) = \frac{1}{4\alpha^2}[\sqrt{(4(n+1)(n+2))}\psi_{n+2} + 2(n+1)\psi_n + 2n\psi_n + \sqrt{4n(n-1)}\psi_{n-2}]$

$= \frac{1}{2\alpha^2}[\sqrt{(n+1)(n+2)}\psi_{n+2} + (2n+1)\psi_n + \sqrt{n(n-1)}\psi_{n-2}]$

(3)

$\bar{x} = \int \psi_n^* x\psi_n = 0$

$V = \frac{1}{2}Kx^2; \bar{V} = \int \psi_n^* V\psi_n = \frac{1}{2}K\int \psi_n^* x^2\psi_n = \frac{1}{2}K\frac{1}{2\alpha^2}(2n+1) = \frac{K(2n+1)}{4\alpha^2} = \frac{\omega(2n+1)\bar{h}}{4} = \frac{E_n}{2}$

2.8

$H_n'(\alpha z) = 2n\alpha H_{n-1}(\alpha z)$

$\psi_n' = A_n e^{-\alpha^2x^2/2} 2n\alpha H_{n-1}(\alpha x) - \alpha^2 x A_n e^{-\alpha^2x^2/2}[\frac{1}{2\alpha x}(H_{n+1} + 2nH_{n-1})]$
$=2n\alpha A_n e^{-\alpha^2x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2x^2/2} H_{n+1} - \alpha n A_n e^{-\alpha^2x^2/2} H_{n-1}$
$=\alpha n A_n e^{-\alpha^2x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2x^2/2} H_{n+1}$
$A_n = \frac{1}{\sqrt{2n}} A_{n-1}; A_n = \sqrt{2n+2} A_{n+1}$
$=\alpha\sqrt{\frac{n}{2}}\psi_{n-1} - \alpha\sqrt{\frac{n+1}{2}}\psi_{n+1}$
(2)
$\frac{d}{dx}\psi_n' = \alpha[\sqrt{\frac{n}{2}}\psi_{n-1}' - \sqrt{\frac{n+1}{2}}\psi_{n+1}']$
$\frac{d}{dx}\psi_n' = \alpha[\sqrt{\frac{n}{2}}[\alpha\sqrt{\frac{n-1}{2}}\psi_{n-2} - \alpha\sqrt{\frac{n}{2}}\psi_n] - \sqrt{\frac{n+1}{2}}[\alpha\sqrt{\frac{n+1}{2}}\psi_n] - \alpha\sqrt{\frac{n+2}{2}}\psi_{n+2}]$
$\frac{d}{dx}\psi_n' = \alpha^2/2[\sqrt{(n-1)n}\psi_{n-2} - n\psi_n - (n+1)\psi_n + \sqrt{(n+1)(n+2)}\psi_{n+2}]$
$\frac{d}{dx}\psi_n' = \alpha^2/2[\sqrt{(n-1)n}\psi_{n-2} - (2n+1)\psi_n + \sqrt{(n+1)(n+2)}\psi_{n+2}]$
(3)
$\bar{p} = \int \psi_n^* p \psi_n = 0$
$\bar{T} = \int \psi_n^* T \psi_n = \int \psi_n^*(-\bar{h}^2/(2m))\frac{d^2}{dx^2}\psi_n = \frac{-\bar{h}^2}{2m}\int \psi_n^* \frac{d^2}{dx^2}\psi_n = \frac{-\bar{h}^2}{2m}[-\frac{\alpha^2}{2}(2n+1)] = \frac{\bar{h}\alpha^2}{4m}(2n+1) = \frac{\bar{h}\omega}{4}(2n+1) = \frac{1}{2}E_n$
2.9
$\Delta x \Delta p = \sqrt{\bar{x^2}\bar{p^2}} = \sqrt{E_n m E_n / K} = \frac{E_n}{\omega} = \frac{\bar{h}}{2}(2n+1)$
3.14
l_zEigenstate:$\psi_m = \sqrt{\frac{1}{2\pi}} e^{im\phi}$
$\int \phi^* l_z \phi = \int (l_z\phi)^* \phi$
$\int \phi^*[l_y, l_z]\phi = \int \phi^* l_y l_z \phi - \phi^* l_z l_y \phi = \int \phi^* l_y \phi \bar{h} m - (l_z\phi)^* l_y \phi = \bar{h} m \bar{l_y} - \bar{h} m \bar{l_y} = 0 = i\bar{h} l_x$
$\int \phi^*[l_z, l_x]\phi = \int \phi^* l_z l_x \phi - \phi^* l_x l_z \phi = \int (l_z\phi)^* l_x \phi - (l_x\phi)^* l_z \phi = \int \phi^* \bar{h} m l_x \phi - \bar{h} m \phi^* l_x \phi = 0 = i\bar{h} l_y$

5 Nov 4 HW

3.15
$\psi = Y_{lm}$ so $\bar{l_x^2} = \int \psi^* l_x^2 \psi d\Omega$ and $l_x^2 = \frac{1}{i\bar{h}}[l_y, l_z] l_x = \frac{1}{i\bar{h}}(l_y l_z - l_z l_y) l_x = \frac{1}{i\bar{h}}(l_y l_z l_x - l_z l_y l_x)$ and we know $l_z l_x = l_x l_z + i\bar{h} l_y$ so :$\frac{1}{i\bar{h}}(l_y l_x l_z + i\bar{h} l_y^2 - l_z l_y l_x) = \frac{1}{i\bar{h}} l_y l_x l_z + l_y^2 - \frac{1}{i\bar{h}} l_z l_y l_x = l_x^2$
so $\bar{l_x^2} = \int \psi^*(\frac{1}{i\bar{h}} l_y l_x l_z + l_y^2 - \frac{1}{i\bar{h}} l_z l_y l_x)\psi d\Omega = \bar{l_y^2} + \int \psi^* \frac{1}{i\bar{h}} l_t l_x \bar{h} m \psi - \frac{1}{i\bar{h}} \bar{h} m \psi^* l_y l_x \psi d\Omega = \bar{l_y^2}$
so we have $\bar{l_x^2} = \bar{l_y^2}$ now
we know $l^2 = l_x^2 + l_y^2 + l_z^2$,so $\bar{l^2} = 2\bar{l_x^2} + \bar{l_z^2}$
$\bar{l_x^2} = \frac{1}{2}(\int \psi^* l^2 \psi - \psi^* l_z^2 \psi d\Omega) = \frac{1}{2}(l(l+1)\bar{h}^2 - m^2\bar{h}^2)$
so $(\Delta\bar{l_x})^2$ we have :$\Delta l_x = \sqrt{-\bar{l_x}^2 + \bar{l_x^2}} = \sqrt{\bar{l_x^2}}$ and thus $(\Delta l_x)^2 = |\bar{l_x^2}| = \frac{1}{2}(l(l+1)\bar{h}^2 - m^2\bar{h}^2)$ so we have
$(\Delta\bar{l_x})^2 = \frac{1}{2}(l(l+1)\bar{h}^2 - m^2\bar{h}^2) = (\Delta\bar{l_y})^2$
3.16
set $\psi = c_1 Y_{11} + c_2 Y_{20}$

$l_z\psi = c_1\bar{h}Y_{11} + c_2 \times 0 = c_1\bar{h}Y_{11}$ so the posibility of status $0\bar{h}$ is $|c_2|^2$,and $\bar{h}$ is 1.so $l_z = \bar{h}$ and the average $=|c_1|^2\bar{h}$

$l^2\psi = c_1 \times 1 \times 2Y_{11}\bar{h}^2 + c_2 \times 2 \times 3Y_{20}\bar{h}^2$ so l^2 can be $2\bar{h}^2$posibility is $|c_1|^2$,and $6\bar{h}^2$ posibility is $|c_2|^2$.

6 Nov 6 HW

*3.18

$e^{-aA}Be^{aA} = \Sigma\frac{-a^nA^n}{n!}B\Sigma\frac{a^mA^m}{m!} = \Sigma\Sigma\frac{-a^nA^n}{n!}B\frac{a^mA^m}{m!} = \Sigma_{m,n}\frac{-1^na^{m+n}A^nBA^m}{n!m!}$

$\Sigma_{m,n}\frac{-1^na^{m+n}A^nBA^m}{n!m!} = B - aAB + aBA - a^2ABA + a^2BA^2/2 + a^2A^2B/2...$

$=B - a[A,B] - \frac{1}{2}a^2(A[A,B])...$ for any m+n=i we can have

$a^i\Sigma_{m+n=i}\frac{-1^nA^nBA^m}{n!m!}$ for i=3 we can have:$a^3\Sigma_{m+n=3}\frac{-1^nA^nBA^m}{n!m!}$

$= a^3[[A,[A,B]],A]/6 = -\frac{a^3}{3!}[A,[A,[A,B]]]$

Now set for i = 0,1,2,3 we set $F_i = \frac{-a}{i+1}[A,F_{i-1}] = \Sigma_{m+n=i}\frac{a^i(-1)^nA^nBA^m}{m!n!}$

Now consider for ant k satisfied F_k satisfied :$F_k = \frac{a^k}{k!}(-1)^k[A,[A,[A,...[A,B]]]] = \Sigma_{m+n=k}\frac{a^k(-1)^nA^nBA^m}{m!n!}$ and for k+1 we have:

$-a/(k+1)[A,F_k] = \frac{a^{k+1}}{(k+1)!}(-1)^{k+1}[A,[A,...[A,B]]] = \Sigma_{m+n=k}\frac{a^{k+1}(-1)^n[A,A^nBA^m]}{m!n!(k+1)}$

and for $[A,A^nBA^m] = [A,A^n]BA^m + A^n[A,BA^m] = [A,A^n]BA^m + A^n[A,B]A^m + A^nB[A,A^m] = A^n[A,B]A^m = A^{n+1}BA^m - A^nBA^{m+1} = A^{n+1}BA^m + (-1)A^nBA^{m+1}$so

$\Sigma_{m+n=k}\frac{a^{k+1}(-1)^n(A^{n+1}BA^m + (-1)A^nBA^{m+1})}{m!n!(k+1)}$

$\Sigma_{m+n=k}\frac{a^{k+1}(-1)^{n+1}A^{n+1}BA^m + a^{k+1}(-1)^nA^nBA^{m+1}}{m!n!(k+1)} = \Sigma_{m+n'=k+1}\frac{a^{k+1}(-1)^{n'}A^{n'}BA^m}{m!(n'-1)!(k+1)} + \Sigma_{m'+n=k+1}\frac{a^{k+1}(-1)^nA^nBA^{m'}}{(m'-1)!n!(k+1)} - \frac{-a^{k+1}ABA^k}{(k+1)!} - \frac{a^{k+1}(-1)^kA^kBA}{(k+1)!} = \Sigma_{m+n'=k+1}n'\frac{a^{k+1}(-1)^{n'}A^{n'}BA^m}{m!(n')!(k+1)} + \Sigma_{m'+n=k+1}m'\frac{a^{k+1}(-1)^nA^nBA^{m'}}{(m')!n!(k+1)} + \frac{a^{k+1}ABA^k}{(k+1)!} - \frac{a^{k+1}(-1)^kA^kBA}{(k+1)!}$ 1 and 3 combine can have a $m\neq$ 1term and 2 and 4 combine can have a $n \neq 1$ term :$\Sigma_{m+n'=k+1;m\neq1}n'\frac{a^{k+1}(-1)^{n'}A^{n'}BA^m}{m!(n')!(k+1)} + \Sigma_{m'+n=k+1;n\neq1}m'\frac{a^{k+1}(-1)^nA^nBA^{m'}}{(m')!n!(k+1)}$ so we get satisfied for full k+1 terms : m=0,1,2...,k total k+1,and we set m+n=k+1,we get:$\Sigma\frac{a^{k+1}(-1)^nA^nBA^m}{m!n!}$

PROOF DONE.

7 Nov 13

4.4: proof:$-\hbar^2\frac{d^2}{dt^2}\bar{A} = \overline{[[A,H],H]}$

PROOF:$\frac{d}{dt}\bar{C} = \frac{1}{i\hbar}\overline{[C,B]}$ and let $\bar{C} = \frac{d}{dt}\bar{A} = \frac{1}{i\hbar}\overline{[A,H]}$ so

$\frac{d}{dt}\frac{d}{dt}\bar{A} = -\frac{1}{\hbar^2}\overline{[[A,H],H]}$

PROVED

4.5: proof:$\frac{d}{dt}\bar{A} = 0$ $\bar{A}$is under a stationary state.

PROVE:$\frac{d}{dt}\bar{A} = \frac{1}{i\hbar}\overline{[A,H]} = \frac{1}{i\hbar}\int\psi^*AH\psi - \psi^*HA\psi = \frac{1}{i\hbar}(E\int\psi^*A\psi - E\psi^*A\psi) = 0$

DONE

8 Nov 18

homework:

4.2:

(a)

$$H = h(q_1) + h(q_2)$$
$$q_1, q_2 = \phi_1, \phi_1; \phi_1, \phi_2; \phi_1, \phi_3; \phi_2, \phi_2; \phi_2, \phi_3; \phi_3, \phi_3.$$

6 kinds

(b)

$$H = h(q_1) + h(q_2)$$
$$q_1, q_2 = \phi_1, \phi_2, \phi_2, \phi_1; \phi_1, \phi_3; \phi_3, \phi_1; \phi_2, \phi_3; \phi_3, \phi_2.$$

3 kinds

(c)

$$3 \times 3 = 9$$

9 kinds

4.3:

(a)

$$3^3 = 27$$

27 kinds

(b)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3} :$$
$$1, 2, 3$$

1 kind

(c)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3} :$$
$$1, 1, 1; 1, 1, 2; 1, 1, 3; 1, 2, 2; 1, 2, 3;$$
$$1, 3, 3; 2, 2, 2; 2, 2, 3; 2, 3, 3; 3, 3, 3;$$

10 kinds

4.8:

proof:

$$(\frac{dF}{dt})_{nn'} = i\omega_{nn'}F_{nn'}, \omega_{nn'} = (E_n - E_{n'})/\hbar \tag{1}$$

prove:

$$\frac{dF}{dt} = \frac{1}{i\hbar}[F(t), H]$$

$$(\frac{dF}{dt})_{nn'} = \frac{1}{i\hbar}\int \phi_n F H \phi_{n'} - \phi_n H F \phi_{n'}$$

$$= \frac{1}{i\hbar}(E_{n'}\int \phi_n F \phi_{n'} - E_n \phi_n F \phi_{n'})$$

$$= \frac{1}{i\hbar}(E_{n'} - E_n)\int \phi_n F \phi_{n'}$$

$$= -\frac{1}{i\hbar}\hbar\omega_{nn'}F_{nn'}$$

$$= i\omega_{nn'}F_{nn'}$$

9 Nov 20

4.7: proof:

$$\frac{\partial E_n}{\partial \lambda} =< \psi_n | \frac{\partial H}{\partial \lambda} | \psi_n > \tag{2}$$

PROVE:

$$\begin{aligned}
&E_n \psi_n = \hat{H} \psi_n \\
&\psi_{na}^* E_n \psi_n^a = E_n = \psi_{na}^* \hat{H} \psi_n^a \\
&\frac{\partial E_n}{\partial \lambda} = \frac{\partial}{\partial \lambda} (\psi_{na}^* \hat{H} \psi_n^a) \\
&= \psi_{na}^* \frac{\partial \hat{H}}{\partial \lambda} \psi_n^a =< \psi_n | \frac{\partial \hat{H}}{\partial \lambda} | \psi_n >
\end{aligned}$$

PROVED

10 Nov 25 NOT THIS !!!!!HW is untill page 10

$\psi = c_1 Y_{11} + c_2 Y_{20}$ try to solve all result and probablility of $\hat{l}_x \psi$

SOLVE:

$$\psi = 0 \times Y_{1-1} + 0 \times Y_{10} + c_1 Y_{11} + 0 \times Y_{2-2} + 0 \times Y_{2-1} + c_1 Y_{20} + 0 \times Y_{21} + 0 \times Y_{22}$$
$$\psi^a = \psi^i \phi_i^a$$
$$l_x{}_b^a = \frac{1}{2} l_-{}_b^a + \frac{1}{2} l_+{}_b^a$$
$$l_x{}_j^i = \phi_a^{*i} l_x{}_b^a \phi_j^b$$

for l =1

$$Y_{1-1} l_X Y_{1-1} = Y_{1-1} \frac{1}{2}(l_+ + l_-) Y_{1-1} = \frac{1}{2}(Y_{1-1} a_{1-1} Y_{10} + Y_{1-1} b_{1-1} \times 0) = 0$$
$$Y_{1-1} l_x Y_{10} = \frac{1}{2}(Y_{1-1} a_{10} Y_{11} + Y_{1-1} b_{10} Y_{1-1}) = \frac{1}{2}\sqrt{l(l+1) - m(m-1)} = \frac{1}{2}\sqrt{2}$$
$$Y_{1-1} l_x Y_{11} = 0$$
$$Y_{10} l_x Y_{1-1} = \frac{1}{2}(Y_{10} a_{1-1} Y10) = \frac{1}{2}\sqrt{l(l+1) - m(m+1)} = \frac{1}{2}\sqrt{2}$$
$$Y_{10} l_x Y_{10} = 0$$

$Y_{10}l_xY_{11} = \frac{1}{2}(Y_{10}a_{11} \times 0 + Y_{10}b_{11}Y_{10}) = \frac{1}{2}\sqrt{2}$

$Y_{11}l_xY_{1-1} = 0$

$Y_{11}l_xY_{10} = \frac{1}{2}(Y_{11}a_{10}Y_{11}) = \frac{1}{2}\sqrt{2}$

for any l=2

$Y_{2-2}l_xY_{2-2} = 0$

$Y_{2-2}l_xY_{2-1} = \frac{1}{2}Y_{2-2}b_{2-1}Y2-2 = \frac{1}{2}\sqrt{6-(-1)(-2)} = 1$

$Y_{2-2}l_xY_{20} = 0$

$Y_{2-1}l_xY_{2-2} = \frac{1}{2}(Y_{2-1}a_{2-2}Y_{2-1}) = \frac{1}{2}\sqrt{6-(-2)(-1)} = 1$

$Y_{2-1}l_xY_{20} = \frac{1}{2}(Y_{2-1}b_{20}Y_{2-1}) = \frac{1}{2}\sqrt{6} = \frac{\sqrt{6}}{2}$

$Y_{20}l_xY_{2-1} = \frac{1}{2}(Y_{20}a_{2-1}Y_{20}) = \frac{1}{2}\sqrt{6} = \frac{\sqrt{6}}{2}$

$Y_{20}l_xY_{21} = \frac{1}{2}Y_{20}b_{21}Y_{20} = \frac{1}{2}\sqrt{6} = \frac{\sqrt{6}}{2}$

$Y_{21}l_xY_{20} = \frac{1}{2}(Y_{21}a_{20}Y_{21}) = \frac{1}{2}\sqrt{6}$

$Y_{21}l_xY_{22} = \frac{1}{2}(Y_{21}b_{22}Y_{21}) = \frac{1}{2}\sqrt{6-2} = 1$

$Y_{22}l_xY_{21} = \frac{1}{2}(Y_{22}a_{21}Y_{22}) = \frac{1}{2}\sqrt{6-2} = 1$

we have :

$l_{x}{}^{1}_{2} = l_{x}{}^{2}_{1} = l_{x}{}^{2}_{3} = l_{x}{}^{3}_{2} = \frac{1}{\sqrt{2}}\hbar$

$l_{x}{}^{4}_{5} = l_{x}{}^{5}_{4} = l_{x}{}^{7}_{8} = l_{x}{}^{8}_{7} = \hbar$

$l_{x}{}^{5}_{6} = l_{x}{}^{6}_{5} = l_{x}{}^{6}_{7} = l_{x}{}^{7}_{6} = \frac{\sqrt{6}}{2}\hbar$

$$l_{x}{}^{i}_{j} = \begin{bmatrix} 0 & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & 0 & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & -\lambda & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & -\lambda \end{vmatrix} = 0$$

$\lambda(\lambda^2 - \frac{\hbar^2}{2}) - \frac{\hbar^2}{2}\lambda = 0$

$\lambda_1 = 0; \lambda_2 = \hbar; \lambda_3 = -\hbar$

so eigenstate :ξ

$$\begin{bmatrix} 0 & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & 0 & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & 0 \end{bmatrix} \times \vec{\xi_0} = 0; \xi_0^2 = 0; \xi_0^1 = -\xi_0^3;$$

$$\vec{\xi_0} = \begin{bmatrix} k \\ 0 \\ -k \end{bmatrix}$$

$$\begin{bmatrix} \hbar & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & \hbar & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & \hbar \end{bmatrix} \times \vec{\xi}_{-1} = 0; \xi^1_{-1} + \frac{1}{\sqrt{2}}\xi^2_{-1} = 0; \frac{1}{\sqrt{2}}\xi^2_{-1} + \xi^3_{-1} = 0; \xi^1_{-1} = \xi^3_{-1} = -\frac{1}{\sqrt{2}}\xi^2_{-1}$$

$$\vec{\xi}_{-1} = \begin{bmatrix} k \\ -\sqrt{2}k \\ k \end{bmatrix}$$

$$\begin{bmatrix} -\hbar & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & -\hbar & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & -\hbar \end{bmatrix} \times \vec{\xi_1} = 0; \xi_1^1 = \frac{1}{\sqrt{2}}\xi_1^2; \frac{1}{\sqrt{2}}\xi_1^2 = \xi_1^3; \xi_1^1 = \xi_1^3 = \frac{1}{\sqrt{2}}\xi_1^2$$

$$\vec{\xi_1} = \begin{bmatrix} k \\ \sqrt{2}k \\ k \end{bmatrix}$$

the posibality:

$\phi^a = \frac{1}{\sqrt{2}}Y_{1-1} + \frac{1}{\sqrt{2}}Y_{10} = \frac{1}{\sqrt{2}}\phi_1^a + \frac{1}{\sqrt{2}}\phi_2^a$

$P_0 = (\phi_a^{*3}\xi_0^a)^2 = \frac{1}{4}$

$P_{-1} = (\phi_a^{*3}\xi_{-1}^a)^2 = (\frac{1}{2\sqrt{2}} - \frac{1}{2})^2$

$P_1 = (\phi_a^{*3}\xi_1^a)^2 = (\frac{1}{2\sqrt{2}} + \frac{1}{2})^2$

now consider the l=2

$$\begin{bmatrix} 0 & \hbar & 0 & 0 & 0 \\ \hbar & 0 & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \hbar \\ 0 & 0 & 0 & \hbar & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & \hbar & 0 & 0 & 0 \\ \hbar & -\lambda & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -\lambda & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & -\lambda & \hbar \\ 0 & 0 & 0 & \hbar & -\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -\lambda & 0 & 0 & 0 & 0 \\ \hbar & \frac{\hbar^2}{\lambda}-\lambda & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -\lambda & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & \frac{\hbar^2}{\lambda}-\lambda & \hbar \\ 0 & 0 & 0 & 0 & -\lambda \end{vmatrix} = 0$$

$$= \lambda^2 \begin{vmatrix} \frac{\hbar^2}{\lambda}-\lambda & \frac{\sqrt{6}\hbar}{2} & 0 \\ \frac{\sqrt{6}\hbar}{2} & -\lambda & \frac{\sqrt{6}\hbar}{2} \\ 0 & \frac{\sqrt{6}\hbar}{2} & \frac{\hbar^2}{\lambda}-\lambda \end{vmatrix} = \lambda^2[(\frac{\hbar^2}{\lambda}-\lambda)(-\hbar^2+\lambda^2-\frac{3}{2}\hbar^2) - \frac{\sqrt{6}\hbar}{2}(\frac{\sqrt{6}\hbar}{2}\frac{\hbar^2}{\lambda} - \frac{\sqrt{6}\hbar}{2}\lambda)]$$

$$= \lambda^2[(\frac{\hbar^2}{\lambda}-\lambda)(\lambda^2-\frac{5}{2}\hbar^2) - \frac{3}{2}\frac{\hbar^4}{\lambda} + \frac{3}{2}\hbar^2\lambda] = \lambda^2[(\lambda\hbar^2 - \frac{5}{2}\frac{\hbar^4}{\lambda} - \lambda^3 + \frac{5}{2}\hbar^2\lambda) - \frac{3}{2}\frac{\hbar^4}{\lambda} + \frac{3}{2}\hbar^2\lambda]$$

$$= \lambda^2(5\lambda\hbar^2 - 4\frac{\hbar^4}{\lambda} - \lambda^3)$$

$\lambda_1 = 0; \lambda_2 = \hbar; \lambda_3 = -\hbar; \lambda_4 = 2\hbar; \lambda_5 = -2\hbar$

$$\begin{bmatrix} -\hbar & \hbar & 0 & 0 & 0 \\ \hbar & -\hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -\hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & -\hbar & \hbar \\ 0 & 0 & 0 & \hbar & -\hbar \end{bmatrix} \times \vec{\xi_1} = 0$$

$$\begin{bmatrix} \hbar & \hbar & 0 & 0 & 0 \\ \hbar & \hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & \hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & \hbar & \hbar \\ 0 & 0 & 0 & \hbar & \hbar \end{bmatrix} \times \vec{\xi_{-1}} = 0$$

$$\begin{bmatrix} 2\hbar & \hbar & 0 & 0 & 0 \\ \hbar & 2\hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & 2\hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & 2\hbar & \hbar \\ 0 & 0 & 0 & \hbar & 2\hbar \end{bmatrix} \times \vec{\xi_{-2}} = 0$$

$$\begin{bmatrix} -2\hbar & \hbar & 0 & 0 & 0 \\ \hbar & -2\hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -2\hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & -2\hbar & \hbar \\ 0 & 0 & 0 & \hbar & -2\hbar \end{bmatrix} \times \vec{\xi}_2 = 0$$

$$\begin{bmatrix} 0 & \hbar & 0 & 0 & 0 \\ \hbar & 0 & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \hbar \\ 0 & 0 & 0 & \hbar & 0 \end{bmatrix} \times \vec{\xi}_0 = 0$$

$$\vec{\xi}_{-2} = \frac{1}{4}\begin{bmatrix} 1 \\ 2 \\ -\sqrt{6} \\ 2 \\ 1 \end{bmatrix}; \vec{\xi}_{-1} = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ -1 \end{bmatrix}; \vec{\xi}_0 = \sqrt{\frac{3}{8}}\begin{bmatrix} 1 \\ 0 \\ -\sqrt{\frac{2}{3}} \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{\xi}_1 = \frac{1}{2}\begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}; \vec{\xi}_2 = \frac{1}{4}\begin{bmatrix} 1 \\ -2 \\ \sqrt{6} \\ -2 \\ 1 \end{bmatrix}$$

$$P_0 = \frac{1}{4}; P_1 = 0; P_2 = \frac{3}{8}; P_{-2} = \frac{3}{8}; P_{-1} = 0$$

SOLVED(WHY SO FUCKING LOOOOOOOOONG?????)

11 Nov 27 HW

homework

7.1:

IF:$A^2 = B^2 = C^2 = 1; BC - CB = iA.$

PROVE:(a)AB+BA=AC+CA=0;(b)B,C in image of A.

PROVE:

(a):

$$iAB + iBA = BCB - CBB + BBC - BCB = BBC - CBB = 0$$
$$iAC + iCA = BCC - CBC + CBC - CCB = 0$$

(b)/home/yuri/books/output/1/1758960262251.jpg

$$AB + BA = 0; \phi_i^* AB\phi_i + \phi^* iBA\phi_j = 0$$
$$\lambda_I \phi_i^* B\phi^j + \phi^{*i} B\lambda_J \phi_j = \lambda_I B_j^i + \lambda_J B_j^i = (\lambda_I + \lambda_J)B_j^i = 0;$$
$$B^2 = 1 \neq 0, B_i^i = 0, B_j^i \neq 0; \lambda_I + \lambda_J = 0; \lambda_1 = a; \lambda_2 = -a$$
$$A = \begin{bmatrix} a & 0 \\ 0 & -a \end{bmatrix}; B = \begin{bmatrix} 0 & b_1 \\ b_2 & 0 \end{bmatrix}; C = \begin{bmatrix} 0 & c_1 \\ c_2 & 0 \end{bmatrix}$$
$$BB = \begin{bmatrix} b_1 b_2 & 0 \\ 0 & b_1 b_2 \end{bmatrix}; b_1 = \frac{1}{b_2}; B = \begin{bmatrix} 0 & b \\ \frac{1}{b} & 0 \end{bmatrix}; C = \begin{bmatrix} 0 & c \\ \frac{1}{c} & 0 \end{bmatrix}$$
$$BC = \begin{bmatrix} \frac{b}{c} & 0 \\ 0 & \frac{c}{b} \end{bmatrix}; CB = \begin{bmatrix} \frac{c}{b} & 0 \\ 0 & \frac{b}{c} \end{bmatrix}$$
$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}; B = \begin{bmatrix} 0 & b \\ \frac{1}{b} & 0 \end{bmatrix}; C = \begin{bmatrix} 0 & c \\ \frac{1}{c} & 0 \end{bmatrix}$$
$$BC - CB = iA; ia = \frac{b}{c} - \frac{c}{b};$$

7.2:

$$A^2 = 0, AA^\dagger + A^\dagger A = 1.B = A^\dagger A.$$

PROVE:(a)$B^2 = B$;(b)in image of B to solve A.

(a)

$$B^2 = A^\dagger AA^\dagger A = A^\dagger A(1 - AA^\dagger) = A^\dagger A = B$$

PROVED

(b)

$B^i_j = diag(\lambda_1, \lambda_2)$
$B^2 = B; B = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}; \lambda_1^2 = \lambda_1, \lambda_2^2 = \lambda_2$
WLOG set $:B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; A^\dagger A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, AA^\dagger = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}; A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; A^2 = \begin{bmatrix} a^2+bc & ab+bd \\ ca+bd & cb+d^2 \end{bmatrix} = 0; a = d$
$bb^* = 0, cc^* = 1; b = 0, c = \sqrt{-k^2+1} + ki$

12 Dec 3 HW

I:8.1:
(a):

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0; \lambda = \pm 1; \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \times \vec{v} = 0$$
$$\vec{v} = \begin{bmatrix} k \\ k \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$
$$\lambda = -1; \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \times \vec{v}_1 = 0; \vec{v}_1 = \begin{bmatrix} k \\ -k \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}$$

(b):

$$\phi_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}; \phi_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}$$
$$\psi_i^j = \psi_i^a \phi_a^{*j}; \psi_1^1 = \frac{1}{\sqrt{2}}; \psi_1^2 = \frac{1}{\sqrt{2}}; \psi_2^1 = \frac{1}{\sqrt{2}}; \psi_2^2 = \frac{-1}{\sqrt{2}}$$
$$\phi^a = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \times \psi^a$$

(c):

$$S^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$
$$S\sigma_x S^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

II:8.2:

$$\sigma \cdot n = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z = \sigma_x sin\theta cos\phi + \sigma_y sin\theta sin\phi + \sigma_z cos\theta$$
$$\phi^{*i} \sigma \cdot n \phi_j = \sigma_{x}{}^{i}_{j} sin\theta cos\phi + \sigma_{y}{}^{i}_{j} sin\theta sin\phi + \sigma_{z}{}^{i}_{j} cos\theta$$
$$= \begin{bmatrix} cos\theta & sin\theta cos\phi - isin\theta sin\phi \\ sin\theta cos\phi + isin\theta sin\phi & -cos\theta \end{bmatrix}$$
$$= \begin{bmatrix} cos\theta & sin\theta e^{-i\phi} \\ sin\theta e^{i\phi} & -cos\theta \end{bmatrix}$$

III:CALCULATE:$[\sigma_x, \sigma_z^n]$;$[\sigma_x, e^{\sigma_z}]$

$$\sigma_z^n = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}^n = \begin{bmatrix} 1 & 0 \\ 0 & -1^n \end{bmatrix}; \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1^n \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & -1^n \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & -1^n \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -1^n & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & -1^n - 1 \\ 1 + -1^{n+1} & 0 \end{bmatrix}$$

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8.2:

$$
\begin{aligned}
&\sigma \cdot n = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z \\
&\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times sin\theta cos\phi + \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \times sin\theta sin\phi + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times cos\theta \\
&= \begin{bmatrix} cos\theta & sin\theta cos\phi - isin\theta sin\phi \\ sin\theta cos\phi + isin\theta sin\phi & -cos\theta \end{bmatrix} = \begin{bmatrix} cos\theta & sin\theta e^{-i\phi} \\ sin\theta e^{i\phi} & -cos\theta \end{bmatrix}
\end{aligned}
$$

8.3:

$$
\begin{aligned}
&\overline{S_x^2} = \phi^* S_x^2 \phi = \phi^* \frac{1}{4}(S_+ + S_-)(S_+ + S_-)\phi \\
&= [\phi^*(S_+^2 + S_+S_- + S_-S_+ + S_-^2)\phi] = [\phi^*(S_+S_-)\phi] = [\phi^* \frac{\hbar^2}{4}\phi] = \frac{\hbar^2}{4} \\
&\overline{S_x} = \phi^* \frac{1}{2}(S_+ + S_-)\phi = 0; \sqrt{\overline{S_x^2} - 0} = \frac{\hbar}{2} \\
&\overline{(\Delta S_x)}^2 = \frac{\hbar^2}{4} \\
&\overline{S_y^2} = \phi^* - 1 \times \frac{1}{4}(S_+ - S_-)(S_+ - S_-)\phi = -\frac{1}{4}\phi^*(S_+^2 - S_+S_- - S_-S_+ + S_-^2)\phi \\
&= -\frac{1}{4}\phi^*(-\hbar^2\phi) = \frac{\hbar^2}{4} \\
&\overline{S_y} = \phi^* S_y \phi = \frac{1}{2i}\phi^*(S_+ - S_-)\phi = 0 \\
&\overline{\Delta S_y}^2 = (\sqrt{\overline{S_y^2} - \overline{S_y}^2})^2 = \frac{\hbar^2}{4}
\end{aligned}
$$

14 Dec 10

I:Calc:

$$\hat{H} = g\sigma_{x1}\sigma_{x2} \quad ; \quad \psi_0^{ab} = \frac{1}{\sqrt{2}}\chi^a_{\frac{1}{2}}\chi^b_{\frac{1}{2}} + \frac{1}{\sqrt{2}}\chi^a_{\frac{1}{2}}\chi^b_{-\frac{1}{2}}$$

Solve:

use base $\{\chi_{00}, \chi_{10}, \chi_{11}, \chi_{1-1}\} = \{\phi_0, \phi_1, \phi_2, \phi_3\}$

$=\frac{1}{\sqrt{2}}(\chi_1\chi_0 - \chi_0\chi_1); \frac{1}{\sqrt{2}}(\chi_1\chi_0 + \chi_0\chi_1); \chi_1\chi_1; \chi_0\chi_0$

$\hat{H}^i_j = \phi^{*i}_b \hat{H}^b{}_a \phi^a_j$

H^i_j	ϕ_0	ϕ_1	ϕ_2	ϕ_3
ϕ_0^*	$-g$	0	0	0
ϕ_2^*	0	g	0	0
ϕ_3^*	0	0	0	g
ϕ_4^*	0	0	g	0

$\hat{H}^0_0 = \frac{1}{2}\phi_0^* g(\chi_0\chi_1 - \chi_1\chi_0) = \frac{1}{2}g(\chi_1^*\chi_0^* - \chi_0^*\chi_1^*)(\chi_0\chi_1 - \chi_1\chi_0) = -g$

$\hat{H}^0_1 = \frac{1}{2}g\phi_0^*(\chi_0\chi_1 + \chi_1\chi_0) = \frac{1}{2}g(\chi_1^*\chi_0^* - \chi_0^*\chi_1^*)(\chi_0\chi_1 + \chi_1\chi_0) = 0$

$\hat{H}^0_2 = \hat{H}^0_3 = 0$

$\hat{H}^1_1 = \frac{1}{2}g\phi_1^*(\chi_0\chi_1 + \chi_1\chi_0) = \frac{1}{2}g(\chi_1\chi_0 + \chi_0\chi_1)(\chi_0\chi_1 + \chi_1\chi_0) = g$

$\hat{H}^1_2 = \hat{H}^1_3 = 0$

$\hat{H}^2_2 = g\phi_2^*\hat{H}\phi_2 = 0$

$\hat{H}^2_3 = g\phi_2^*\hat{H}\phi_3 = g$

$\hat{H}^3_3 = 0$

$$\begin{vmatrix} -g-\lambda & 0 & 0 & 0 \\ 0 & g-\lambda & 0 & 0 \\ 0 & 0 & -\lambda & g \\ 0 & 0 & g & -\lambda \end{vmatrix} = 0; \quad \text{so } \lambda = (\lambda^2 - g^2)(\lambda^2 - g^2) = 0$$

$v_0^a = \phi_0; v_1^a = \phi_1; v_2^a = \frac{1}{\sqrt{2}}(\phi_2 + \phi_3); v_3^a = \frac{1}{\sqrt{2}}(\phi_2 - \phi_3); \lambda_0 = -g; \lambda_1 = g; \lambda_2 = g$

$; \lambda_3 = -g$

$\psi_0^{ab} = \frac{1}{\sqrt{2}}\phi_2^a + \frac{1}{2}(\phi_0 + \phi_1) = \frac{1}{2}v_0 + \frac{1}{2}v_1 + \frac{1}{2}(v_3 + v_2)$

II:8.8:

$$\psi_0 = \chi_1\chi_0 \text{ and the Hamiltonian } : \hat{H} = A s_1 \cdot s_2.$$

Solve:

$\hat{H} = \frac{A}{4}\hbar^2\sigma_1\sigma_2 = \frac{A}{4}\hbar^2(\sigma_{1x}\sigma_{2x} + \sigma_{1y}\sigma_{2y} + \sigma_{1z}\sigma_{2z})$

bases are :$\chi_{00} = \frac{1}{\sqrt{2}}(\chi_1\chi_0 - \chi_0\chi_1); \chi_{10} = \frac{1}{\sqrt{2}}(\chi_1\chi_0 + \chi_0\chi_1); \chi_{11} = \chi_1\chi_1; \chi_{1-1} = \chi_0\chi_0$

$\hat{H}^i_j = \phi^{*i}_a \hat{H}^a_b \phi^b_j$

$\hat{H}^i_j$	ϕ_0	ϕ_1	ϕ_2	ϕ_3
ϕ^*_0	$-\frac{3}{4}A\hbar^2$	0	0	0
ϕ^*_1	0	$\frac{1}{4}A\hbar^2$	0	0
ϕ^*_2	0	0	$\frac{1}{4}A\hbar^2$	0
ϕ^*_3	0	0	0	$\frac{1}{4}A\hbar^2$

$\hat{H}^a_0 = \frac{A}{4}\hbar^2\frac{3}{\sqrt{2}}(\chi_0\chi_1 - \chi_1\chi_0) = -\frac{3}{4}A\hbar^2\chi_{00}$

$\hat{H}^0_0 = -\frac{3}{4}A\hbar^2$

$\hat{H}^a_1 = \frac{1}{4}A\hbar^2\frac{1}{\sqrt{2}}(\chi_0\chi_1 + \chi_1\chi_0) = \frac{1}{4}A\hbar^2\chi_{10}$

$\hat{H}^1_1 = \frac{1}{4}A\hbar^2$

$\hat{H}^a_2 = \frac{1}{4}A\hbar^2(\chi_0\chi_0 - \chi_0\chi_0 + \chi_1\chi_1) = \frac{1}{4}A\hbar^2\chi_{11}$

$\hat{H}^2_2 = \frac{1}{4}A\hbar^2$

$\hat{H}^a_3 = \frac{1}{4}A\hbar^2(\chi_1\chi_1 - \chi_1\chi_1 + \chi_0\chi_0) = \frac{1}{4}A\hbar^2\chi_{1-1}$

$\hat{H}^3_3 = \frac{1}{4}A\hbar^2$

$\psi_0 = \frac{1}{\sqrt{2}}(\chi_{00} + \chi_{10})$

$\psi_0(t) = \frac{1}{\sqrt{2}}\chi_{00}e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}e^{-i\frac{1}{4}A\hbar t} = \frac{1}{2}e^{i\frac{3}{4}\hbar At}(\chi_1\chi_0 - \chi_0\chi_1) + \frac{1}{2}e^{-i\frac{1}{4}\hbar At}(\chi_1\chi_0 + \chi_0\chi_1)$

$= e^{i\frac{1}{4}\hbar At}[\frac{1}{2}e^{i\frac{1}{2}\hbar At}(\chi_1\chi_0 - \chi_0\chi_1) + \frac{1}{2}e^{-\frac{1}{2}\hbar At}(\chi_1\chi_0 + \chi_0\chi_1)]$

$$= e^{i\frac{1}{4}A\hbar t}[cos(\frac{1}{2}\hbar At)\chi_1\chi_0 + \frac{1}{i}sin(\frac{1}{2}\hbar At)\chi_0\chi_1]$$

$(a):$

$$= cos^2(\frac{1}{2}\hbar At)$$

$(b):$

$$= 0$$

$(c):$

$$= \frac{1}{2}; = \frac{1}{2}$$

$(d):$

$$< s_{1x} >= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})s_{1x}(\frac{1}{\sqrt{2}}\chi_{00}e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}e^{-i\frac{1}{4}A\hbar t}) = 0$$

$$< s_{1y} >= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})s_{1y}(\frac{1}{\sqrt{2}}\chi_{00}e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}e^{-i\frac{1}{4}A\hbar t}) = 0$$

$$< s_{1z} >= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})s_{1z}(\frac{1}{\sqrt{2}}\chi_{00}e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}e^{-i\frac{1}{4}A\hbar t})$$

$$= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})\frac{1}{2}\hbar(\frac{1}{2}e^{i\frac{3}{4}\hbar At}(\chi_1\chi_0 + \chi_0\chi_1) + \frac{1}{2}e^{-i\frac{1}{4}\hbar At}(\chi_1\chi_0 - \chi_0\chi_1))$$

$$= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})\frac{1}{2}\hbar(\frac{1}{\sqrt{2}}e^{i\frac{3}{4}\hbar At}\chi_{10} + \frac{1}{\sqrt{2}}e^{-i\frac{1}{4}\hbar At}\chi_{00})$$

$$= \frac{1}{2}e^{i\hbar At} + \frac{1}{2}e^{-i\hbar At} = \frac{1}{2}\hbar cos(\hbar At)$$

$$< s_{2x} >=< s_{2y} >= 0$$

$$< s_{2z} >= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})s_{2z}(\frac{1}{\sqrt{2}}\chi_{00}e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}e^{-i\frac{1}{4}A\hbar t})$$

$$= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})s_{2z}(\frac{1}{2}e^{i\frac{3}{4}A\hbar t}(\chi_1\chi_0 - \chi_0\chi_1) + \frac{1}{2}e^{-i\frac{1}{4}A\hbar t}(\chi_1\chi_0 + \chi_0\chi_1))$$

$$= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t})s_{2z}(\frac{1}{\sqrt{2}}e^{i\frac{3}{4}A\hbar t}(-\chi_{10}) + \frac{1}{\sqrt{2}}e^{-i\frac{1}{4}A\hbar t}(-\chi_{00}))$$

$$= -\frac{1}{2}\hbar cos(A\hbar t)$$

15 Dec 11 HW

I:8.5:

(a):Prove

$$e^{i\lambda\sigma_z} = cos\lambda + i\sigma_z sin\lambda$$

Proof.

$$e^{i\lambda\sigma_z} = \Sigma\frac{(i\lambda)^n}{n!}\sigma_z^n; cos\lambda = \Sigma\frac{(-1)^n}{2n!}\lambda^{2n}$$
$$sin\lambda = \Sigma\frac{(-1)^n}{2n+1!}\lambda^{2n+1}$$
$$\sigma_z = \begin{bmatrix}1 & 0\\ 0 & (-1)^n\end{bmatrix}; e^{i\lambda\sigma_z} = \Sigma\begin{bmatrix}\frac{(i\lambda)^n}{n!} & 0\\ 0 & \frac{(-i\lambda)^n}{n!}\end{bmatrix}$$
$$cos\lambda + i\sigma_z sin\lambda = \begin{bmatrix}cos\lambda + isin\lambda & 0\\ 0 & cos\lambda - isin\lambda\end{bmatrix}$$
$$= \begin{bmatrix}e^{i\lambda} & 0\\ 0 & e^{-i\lambda}\end{bmatrix} = e^{i\lambda\sigma_z}$$

□

(b):Prove

$$e^{i\sigma\cdot A} = cosA + i\sigma\cdot nsinA, A = An, A = |A|$$

Proof.

$$e^{i\sigma\cdot A} = e^{i\sigma\cdot nA} = \Sigma\frac{(iA)^n}{n!}(\sigma\cdot n)^n$$
$$cosA + i\sigma\cdot nsinA$$
$$\sigma\cdot n = n_x\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} + n_y\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix} + n_z\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = \begin{bmatrix}n_z & n_x - in_y\\ n_x + in_y & -n_z\end{bmatrix}$$
$$-> \begin{bmatrix}cosA + isinAn_z & isinA(n_x - in_y)\\ isinA(n_x + in_y) & cosA - isinAn_z\end{bmatrix}$$
$$e^{i\sigma\cdot nA} = \Sigma\frac{(iA)^n}{n!}\begin{bmatrix}n_z & n_x - in_y\\ n_x + in_y & -n_z\end{bmatrix}^n$$
$$\begin{bmatrix}n_z & n_x - in_y\\ n_x + in_y & -n_z\end{bmatrix}\times\begin{bmatrix}n_z & n_x - in_y\\ n_x + in_y & -n_z\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$
$$= \Sigma\frac{-1^n A^{2n}}{2n!}I + \Sigma\frac{-1^n\times iA^{2n+1}}{2n+1}\begin{bmatrix}n_z & n_x - in_y\\ n_x + in_y & -n_z\end{bmatrix} = cosAI + isinA\begin{bmatrix}n_z & n_x - in_y\\ n_x + in_y & -n_z\end{bmatrix}$$
$$= cosA + i\sigma\cdot sinA$$

□

(c):Prove

$$Tre^{i\sigma\cdot A} = 2cosA$$

Proof.

$$e^{i\sigma\cdot A} = cosAI + isinA \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \text{the trace is :}$$
$$= 2cosA + 0 = 2cosA$$

$\square$

II:8.6:
Prove:

$$e^{i\lambda\sigma_z}\sigma_x e^{-i\lambda\sigma_z} = \sigma_x cos2\lambda - \sigma_y sin2\lambda$$

Proof.

$$(cos\lambda + i\sigma_z sin\lambda)\sigma_x(cos\lambda - i\sigma_z sin\lambda)$$
$$= \begin{bmatrix} cos\lambda + isin\lambda & 0 \\ 0 & cos\lambda - isin\lambda \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} cos\lambda - isin\lambda & 0 \\ 0 & cos\lambda + isin\lambda \end{bmatrix}$$
$$= \begin{bmatrix} cos\lambda + isin\lambda & 0 \\ 0 & cos\lambda - isin\lambda \end{bmatrix} \begin{bmatrix} 0 & cos\lambda + isin\lambda \\ cos\lambda - isin\lambda & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & cos^2\lambda - sin^2\lambda + 2icos\lambda sin\lambda \\ cos^2\lambda - sin^2\lambda - 2icos\lambda sin\lambda & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & e^{i2\lambda} \\ e^{-i2\lambda} & 0 \end{bmatrix}$$
$$\sigma_x cos2\lambda - \sigma_y sin2\lambda = \begin{bmatrix} 0 & cos2\lambda + isin2\lambda \\ cos2\lambda - isin2\lambda & 0 \end{bmatrix} = \begin{bmatrix} 0 & e^{i2\lambda} \\ e^{-2i\lambda} & 0 \end{bmatrix}$$

□

III:Backboard Question:

$$H = \frac{\hbar^2}{4} g\sigma_{1x}\sigma_{2x}; \psi = \frac{1}{\sqrt{2}}[\chi_1\chi_0 + \chi_1\chi_1]$$

Solve.

the eigenstate:$\phi_1 = \frac{1}{\sqrt{2}}[\chi_1\chi_0 + \chi_0\chi_1]; \phi_2 = \frac{1}{\sqrt{2}}[\chi_1\chi_0 - \chi_0\chi_1];$

$\phi_3 = \frac{1}{\sqrt{2}}[\chi_1\chi_1 + \chi_0\chi_0]; \phi_4 = \frac{1}{\sqrt{2}}[\chi_1\chi_1 - \chi_0\chi_0]$

$\lambda_1 = 1; \lambda_2 = -1; \lambda_3 = 1; \lambda_4 = -1$

$\psi_0 = \frac{1}{2}[\phi_1 + \phi_2] + \frac{1}{2}[\phi_3 + \phi_4]$

$\phi(t) = \frac{1}{2}\phi_1 e^{-it/\hbar} + \frac{1}{2}\phi_2 e^{it/\hbar} + \frac{1}{2}\phi_3 e^{-it/\hbar} + \frac{1}{2}\phi_4 e^{it/\hbar}$

$\phi^*(t) = \frac{1}{2}\phi_1^* e^{it/\hbar} + \frac{1}{2}\phi_2^* e^{-it/\hbar} + \frac{1}{2}\phi_3^* e^{it/\hbar} + \frac{1}{2}\phi_4^* e^{-it/\hbar}$

$< \sigma_{x1} >= \phi^*(t)\sigma_z \frac{1}{2}[e^{-it/\hbar}\frac{1}{\sqrt{2}}(\chi_1\chi_0 + \chi_0\chi_1 + \chi_1\chi_1 + \chi_0\chi_0) + e^{it/\hbar}\frac{1}{\sqrt{2}}(\chi_1\chi_0 - \chi_0\chi_1 + \chi_1\chi_1 - \chi_0\chi_0)]$

$= \phi^*(t)\frac{1}{2\sqrt{2}}[e^{-it/\hbar}(\chi_0\chi_0 + \chi_1\chi_1 + \chi_0\chi_1 + \chi_1\chi_0) + e^{it/\hbar}(\chi_0\chi_0 - \chi_1\chi_1 + \chi_0\chi_1 - \chi_1\chi_0)]$

$= \phi^*(t)[e^{-it/\hbar}\frac{1}{2}(\phi_1 + \phi_3) - \frac{1}{2}e^{it/\hbar}(\phi_2 + \phi_4)] = 0$

Solved.

16 Dec 16 HW

I:8.9:

Solve:

$\psi = \chi_1$
$H = \frac{eB\hbar}{2mc}\sigma_x$, eigenstate is :$\phi_1 = \frac{1}{\sqrt{2}}[\chi_1 + \chi_0]; \phi_2 = \frac{1}{\sqrt{2}}[\chi_1 - \chi_0]$
$\lambda_1 = \frac{eB\hbar}{2mc}; \lambda_2 = -\frac{eB\hbar}{2mc}$
$\psi = \frac{1}{\sqrt{2}}(\phi_1 + \phi_2)$
$\psi(t) = \frac{1}{\sqrt{2}}\phi_1 e^{-i\frac{eB}{2mc}t} + \frac{1}{\sqrt{2}}\phi_2 e^{i\frac{eB}{2mc}t}$
$< s_x >= \frac{1}{\sqrt{2}}[\phi_1^* e^{iKt} + \phi_2^* e^{-iKt}]\frac{1}{\sqrt{2}}\frac{\hbar}{2}[\phi_1 e^{-iKt} - \phi_2 e^{-iKt}] = \frac{\hbar}{4}(0) = 0$
$< s_y >= \frac{1}{\sqrt{2}}[\phi_1^* e^{iKt} + \phi_2^* e^{-iKt}]\frac{\hbar}{2}\frac{1}{\sqrt{2}}[-e^{-iKt}i\phi_2 + e^{iKt}i\phi_1] = \frac{\hbar}{4}i(e^{2iKt} - e^{-2iKt}) = \frac{-\hbar}{2}sin(2Kt)$
$< s_z >= \frac{1}{\sqrt{2}}[\phi_1^* e^{iKt} + \phi_2^* e^{-iKt}]\frac{\hbar}{2}\frac{1}{\sqrt{2}}[e^{-iKt}\phi_2 + e^{iKt}\phi_1] = \frac{\hbar}{4}(e^{i2Kt} + e^{-i2Kt}) = \frac{\hbar}{2}cos(2Kt)$

II:8.10

Solve

$$\phi_0 = \frac{\sqrt{\alpha}}{\pi^{\frac{1}{4}}}e^{-\alpha^2 x^2/2}$$
$$\phi_1 = \frac{\sqrt{2\alpha}}{\pi^{\frac{1}{4}}}\alpha x e^{-\alpha^2 x^2/2}$$
$$\phi_2 = \frac{1}{\pi^{\frac{1}{4}}}\sqrt{\frac{\alpha}{2}}(2\alpha^2 x^2 - 1)e^{-\alpha^2 x^2/2}$$

two particles without spin can be (wave function must be symmetry):

$\phi_{02}\phi_{01} : \psi = \frac{\alpha}{\sqrt{\pi}}e^{-\frac{\alpha^2}{2}(x_1^2+x_2^2)}E = \hbar\omega$

$\phi_{11}\phi_{02}, \phi_{01}\phi_{12} : \psi = \frac{\alpha^2}{\sqrt{\pi}}[(x_1 + x_2)e^{-\frac{\alpha^2}{2}(x_1^2+x_2^2)}]E = 2\hbar\omega$

$\phi_{11}\phi_{12}, \phi_{01}\phi_{22}, \phi_{21}\phi_{02} : \psi = \frac{1}{\sqrt{2}}\frac{\alpha}{\sqrt{2\pi}}[2\alpha^2(x_1^2 + x_2^2 - 2)e^{-\frac{\alpha^2}{2}(x_1^2+x_2^2)}];$

$\phi = \frac{2\alpha^3}{\sqrt{\pi}}x_1 x_2 e^{-\frac{\alpha^2}{2}(x_1^2+x_2^2)}E = 3\hbar\omega$

two particle with spin can be (wavefunction must be anti-symmetry):

eigenstate of Hamiltonian:

$\phi_0\chi_{sm_s}$

$\phi_{01}\phi_{02}\chi_{00} : \psi = \frac{\alpha}{\sqrt{\pi}}e^{-\frac{\alpha^2}{2}(x_1^2+x_2^2)}\chi_{00}$

the second class

$\psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{12} + \phi_{11}\phi_{02}]\chi_{00}; \psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{12} - \phi_{11}\phi_{02}]\chi_{1M}$

the third class

$\psi = [\phi_{11}\phi_{12}]\chi_{00}; \psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{22} + \phi_{21}\phi_{02}]\chi_{00}; \psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{22} - \phi_{21}\phi_{02}]\chi_{1M}$

III:5.6

Calculate

(a):

$Pdr|_r = \int |\psi_{rlm}|^2 r^2 sin\theta dr d\theta d\phi = \int R_r^2 Y_{lm}^2 r^2 sin\theta dr d\theta d\phi = R^2 r^2 dr \int |Y_{lm}|^2 sin\theta d\theta d\phi$

$= R^2 r^2 dr = \chi^2 dr$

so when P is the biggest then : $\frac{dP}{dr} = 2\chi\frac{d\chi}{dr} = 0$

$\Rightarrow \frac{d(r^{l+1}e^{-\beta r})}{dr} = (l+1)r^l e^{-\beta r} - \beta r^{l+1}e^{-\beta r} = 0$

$\Rightarrow r = \frac{l+1}{\beta} = \frac{n}{\beta} = n^2$

consider the dimension of legth:$L = \frac{\hbar^2}{\mu e^2} = \alpha$ but the atom have Z electonics so:

$\Rightarrow r = n^2\frac{\alpha}{Z}$

(b):

$$< r >= \int \psi_{rlm}^* r \psi_{rml} d\Omega$$

use the virial theorem $\overline{\frac{p^2}{2m}} = \frac{1}{2}\overline{r \cdot \nabla V}$

$E = T + V = \frac{p^2}{2m} + \frac{1}{r} = \frac{\mu Z^2 e^4}{2\hbar^2 n^2} = \frac{Z^2 e^2}{2\alpha n^2}$

$\overline{T} = -\frac{1}{2}\overline{V}; \overline{E} = \frac{1}{2}\overline{V} \Rightarrow \overline{\frac{1}{r}} = \frac{Z}{\alpha n^2}$

From 5.9 we know an equation:

$$\frac{\lambda+1}{n^2} < r^\lambda > -(2\lambda+1)\frac{a_0}{Z} < r^{\lambda-1} > +\frac{\lambda}{4}[(2l+1)^2-\lambda^2]\frac{a_0^2}{Z^2} < r^{\lambda-2} >= 0$$

so we get :

$$\frac{2}{n^2} < r > -3\frac{a_0}{Z} + \frac{1}{4}[(2l+1)^2-1]\frac{a_0^2}{Z^2} < \frac{1}{r} >= 0$$
$$\Rightarrow < r >= (n^2 + \frac{n}{2})\frac{\alpha}{Z}$$

(c):
As the same put $< r >$ into eq.1 can get

$$< r^2 >= n^2(n+1)(n+1)(\frac{\alpha}{2})^2$$
$$\text{so delta equal to :}(\frac{n^3}{2} + \frac{n^2}{4})^{\frac{1}{2}}\frac{\alpha}{Z}$$

17 HW Dec 19

Prove and Calculate:

From

$$[a, a^\dagger] = 1$$

Calculate:

$$(1) : [a, a^\dagger a]$$
$$(2) : [a, (a^\dagger a)^n]$$

Calcultion:

$(1) : [a, a^\dagger a] = a^\dagger [a, a] + [a, a^\dagger] a = a$

$(2) : [a, (a^\dagger a)^2] = (a^\dagger a)a + a(a^\dagger a)$

$[a, (a^\dagger a)^3] = (a^\dagger a)^2 a + a^\dagger a a a^\dagger a + a(a^\dagger a)^2$

Now set the n-1 satisfied:

$[a, (a^\dagger a)^{n-1}] = (a^\dagger a)^{n-2} a + (a^\dagger a)^{n-3} a a^\dagger a + \cdots + a^\dagger a a (a^\dagger a)^{n-3} + a(a^\dagger a)^{n-2}$

For n we have:

$[a, (a^\dagger a)^n] = (a^\dagger a)^{n-1} a + [a, (a^\dagger a)^{n-1}] a^\dagger a$

$=(a^\dagger a)^{n-1} a + (a^\dagger a)^{n-2} a a^\dagger a + (a^\dagger a)^{n-3} a (a^\dagger a)^2 + \cdots + a(a^\dagger a)^{n-1}$

DONE.

18 HW Dec 23

I:(1)Calculate

$$\psi_0 = |n> \quad \text{calculate} : < \hat{x} > . < \hat{p} >, < \hat{x}^2 >, < \hat{p}^2 >, \Delta x, \Delta p \tag{3}$$

Solve:

$$(a) : \psi^* x \psi$$

$$x_b^a = \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{\omega\mu}}(a + a^\dagger)$$

$$\Rightarrow \psi_0^* \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{\omega\mu}}(\sqrt{n}|n-1> + \sqrt{n+1}|n+1>) = 0$$

$$(b) : \psi_0^* p \psi_0$$

$$\hat{p} = \frac{1}{i\sqrt{2}}(a - a^\dagger)\sqrt{\hbar\omega\mu}$$

$$\Rightarrow \psi_0^* \frac{1}{i\sqrt{2}}\sqrt{\hbar\omega\mu}(\sqrt{n}|n-1> + \sqrt{n+1}|n+1>) = 0$$

$$(c) : x^2 = \frac{1}{2}\frac{\hbar}{\omega\mu}(a^2 + aa^\dagger + a^\dagger a + a^{\dagger 2})$$

$$\psi_0^* \frac{1}{2}\frac{\hbar}{\omega\mu}(\sqrt{n(n-1)}|n-2> + (2n+1)|n> + \sqrt{(n+1)(n+1)}|n+2>)$$

$$= \frac{1}{2}\frac{\hbar}{\omega\mu}(2n+1)$$

$$(d) : p^2 = -\frac{1}{2}(\hbar\omega\mu)(a^2 - aa^\dagger - a^\dagger a + a^{\dagger 2})$$

$$\Rightarrow \psi^*(-\frac{1}{2}\hbar\omega\mu)(\sqrt{n(n-1)}|n-2> - (2n+1)|n> + \sqrt{(n+1)(n+2)}|n+2>)$$

$$= \frac{1}{2}\hbar\omega\mu(2n+1)$$

$$(e) : \Delta x = \sqrt{(\overline{x^2}) - \overline{x}^2} = \sqrt{\frac{1}{2}\frac{\hbar}{\omega\mu}(2n+1)}$$

$$\Delta p = \sqrt{(\overline{p^2}) - \overline{p}^2} = \sqrt{\frac{1}{2}\hbar\omega\mu(2n+1)}$$

(2)**Calculate:**

$$\psi_0 = \frac{1}{\sqrt{2}}(|0> + |1>) \quad \Delta x \text{ and } \Delta p$$

Solve:

$$\begin{aligned}
&\overline{x} = \psi^* \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{\omega\mu}}(a + a^\dagger)\psi \\
&\Rightarrow \psi^* \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{\omega\mu}}\frac{1}{\sqrt{2}}(|0> + |1> + \sqrt{2}|2>) = \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{\omega\mu}}
\end{aligned}$$

$$\begin{aligned}
&\overline{p} = \psi^* \frac{1}{i\sqrt{2}}\sqrt{\hbar\omega\mu}(a - a^\dagger)\psi \\
&\Rightarrow \psi^* \frac{1}{i\sqrt{2}}\sqrt{\hbar\omega\mu}\frac{1}{\sqrt{2}}(|0> - |1> - \sqrt{2}|2>) = 0
\end{aligned}$$

$$\begin{aligned}
&p^2 = -\frac{1}{2}(\hbar\omega\mu)(a^2 - aa^\dagger - a^\dagger a + a^{\dagger 2}) \\
&\Rightarrow \psi^*(-\frac{1}{2}(\hbar\omega\mu))(a^2 - 2\hat{N} - 1 + a^{\dagger 2})\psi \\
&\Rightarrow \psi^*(-\frac{1}{2}\hbar\omega\mu)\frac{1}{\sqrt{2}}(-|0> + \sqrt{2}|2> - 3|1> + \sqrt{6}|3>) \\
&\Rightarrow \frac{1}{2}(-\frac{1}{2}\hbar\omega\mu)(-4) = \hbar\omega\mu
\end{aligned}$$

$$\begin{aligned}
&x^2 = \frac{1}{2}\frac{\hbar}{\omega\mu}(a^2 + aa^\dagger + a^\dagger a + a^{\dagger 2}) \\
&\Rightarrow \psi^*(\frac{1}{2}\frac{\hbar}{\omega\mu})(a^2 + 2\hat{N} + 1 + a^{\dagger 2})\psi \\
&\Rightarrow \psi^*(\frac{\hbar}{2\omega\mu})\frac{1}{\sqrt{2}}(|0> + \sqrt{2}|2> + 2|1> + \sqrt{6}|3> + |1>) \\
&\Rightarrow \frac{1}{2}\frac{\hbar}{2\omega\mu}(4) = \frac{\hbar}{\omega\mu}
\end{aligned}$$

$$\Delta p \Delta x = \sqrt{(\sqrt{(\overline{x^2}) - \overline{x}^2})^2 (\sqrt{(\overline{p^2}) - \overline{p}^2})^2} = \sqrt{(\sqrt{\frac{\hbar}{\omega\mu} - \frac{1}{2}\frac{\hbar}{\omega\mu}})^2 \sqrt{(\hbar\omega\mu - 0)^2}}$$

$$\Rightarrow \sqrt{\frac{\hbar}{2\omega\mu}\hbar\omega\mu} = \frac{1}{\sqrt{2}}\hbar$$